

Climate 5000 Series - Gen 4

24K BTU Single Zone Max Performance System

Ducted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

24K Btu Single Zone Max Performance System - Ducted IDU

Indoor part number		8-733-962-712
Indoor model number		BMS500-AAU024-1AHDXD
Outdoor part number		8-733-962-691
Outdoor model number		BMS500-AAS024-1CSXHD
Efficiency Data		AHRI 210/240 2023
SEER2		19
EER2		12.0
HSPF2 Region IV		11.7
HSPF2 Region V		9.3
COP(47F)		3.81
COP (5F)		2.18
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	5000~28000
	Rated Capacity	24000
	Current	8.8
	Input Power	2000
Heating	Capacity Range	7300~31400
	Rated Capacity (47F)	26000
	Rated Capacity (5F)	24000
	Current	8.8
	Input Power	2000
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 24.90
Max. fuse indoor/outdoor (A)		15.0 / 25.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 620
Input (W)		805 / 2045
Rated current (RLA/A)		4.15 / 9.30
Piping		
Refrigerant precharge (oz/kg)		63.49 / 1.80
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		3/8" / Ø9.52
Gas side (in/mm)		5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.32 / 30
Max. refrigerant pipe length (ft/m)		164.00 / 50
Max. difference in height (ft/m)		82.00 / 25

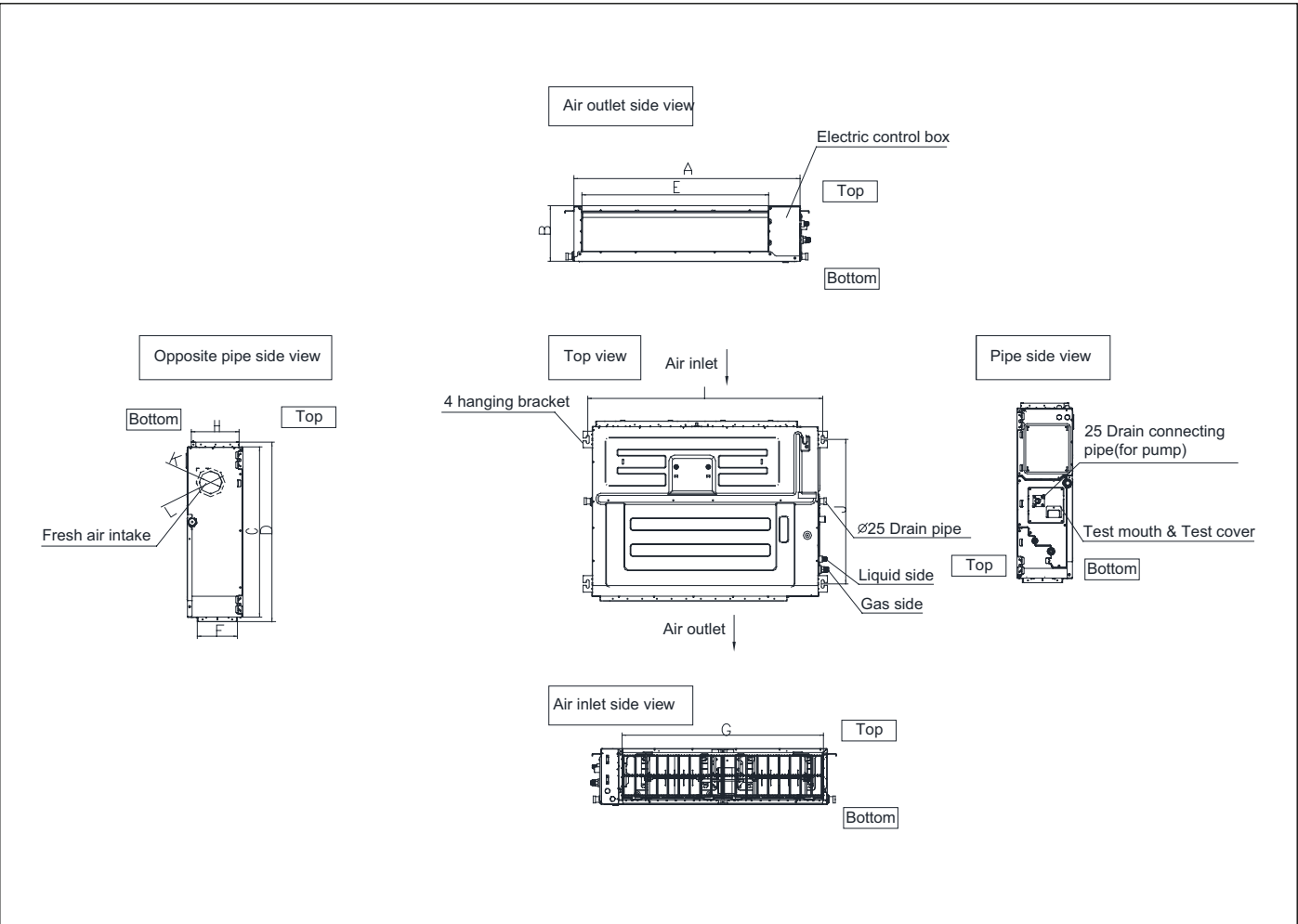
Operation Range

ODU	Cooling (°F / °C)	-22 ~ 122 / -30 ~ 50
	Heating (°F / °C)	-22 ~ 75 / -30 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.5
IDU	Fan motor model	ZKFN-165-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	185.0
	Fan speed, H / M / L (RPM)	920 / 835 / 750
	Air flow, Turbo / H / M / L / Si (CFM)	853.5 / 706.3 / 618.0 / 500.3 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 40.0 / 37.0 / 35 / 29.5
	Net weight (lbs / kg)	80.47 / 36.5
	Gross weight (lbs / kg)	91.93 / 41.7
ODU	ESP range (Pa)	0-200
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-120-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	150
	Fan motor speed, H / / L (RPM)	950 / 800 / 500
	Air flow, Max (CFM)	2236.7
	Noise level, Max (dB(A))	61.0
Accessories	BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
	BMS-DRY-XXD (8733969263)	24V Interface
Accessories	BMS-PK3-7X1 (8733951015)	25ft Lineset

Indoor Unit Dimensions and Connections

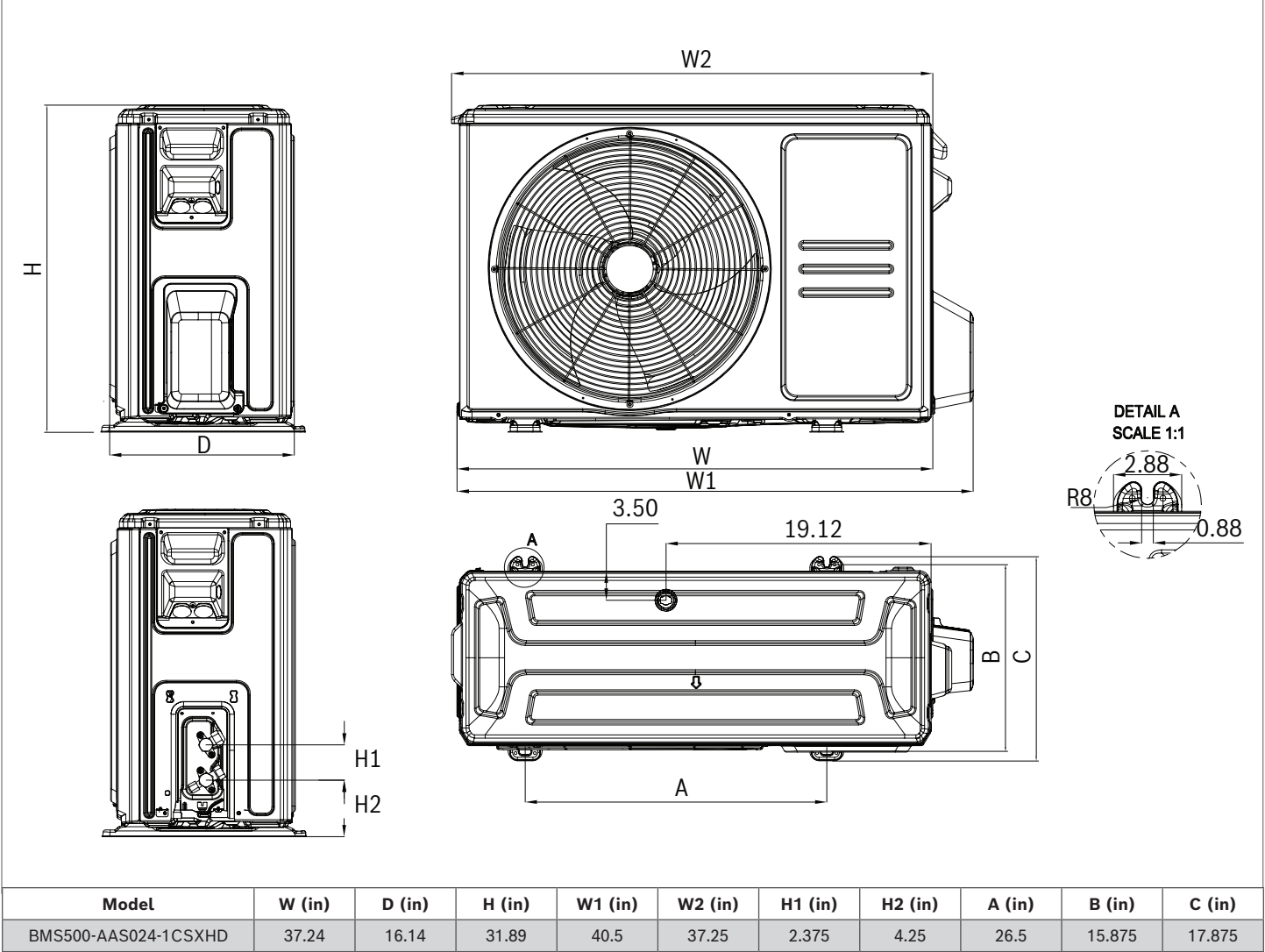


Model	Outline Dimension (in)				Air Outlet Opening Size (in)		Air Return Opening Size (in)		Size of Mounted Lug (in)		Fresh Air Intake Opening Size (in)	
	A	B	C	D	E	F	G	H	I	J	K	L
BMS500-AAU024-1AHDXD	27.6	9.6	29.5	31.3	32.6	7.0	35.1	8.3	40.9	25.2	3.9	5.0

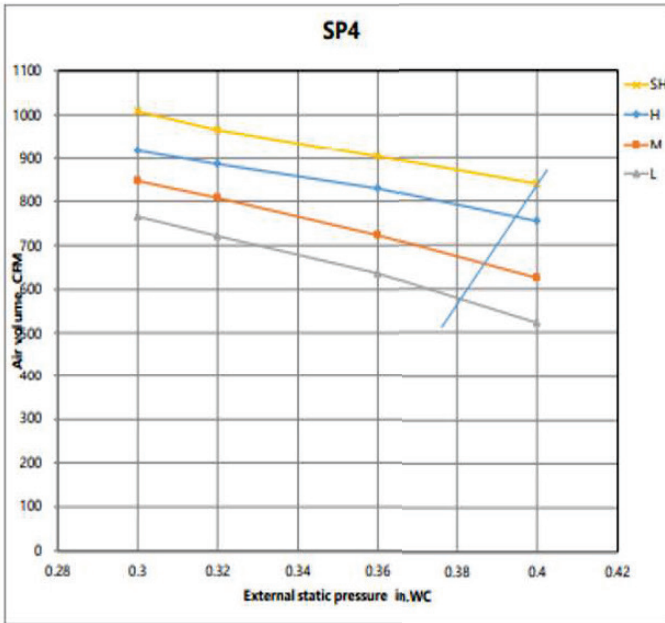
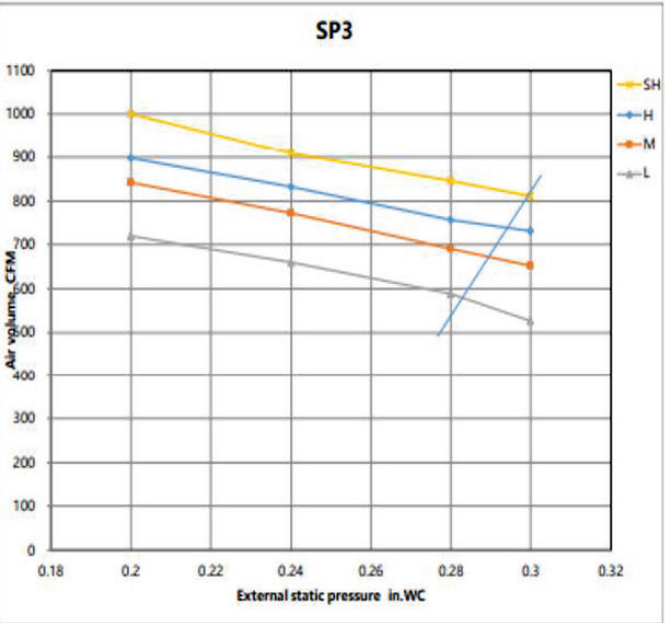
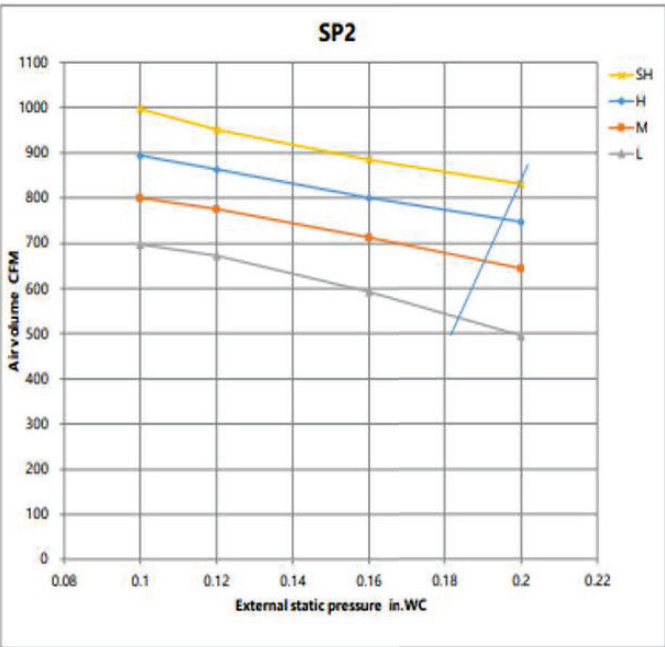
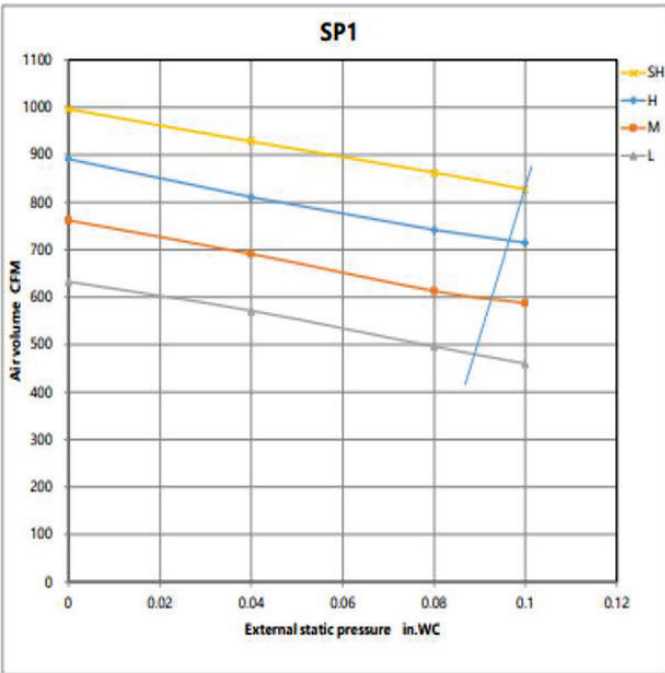
Climate 5000 Series - Gen 4
24K Btu Single Zone Max Performance System - Ducted IDU



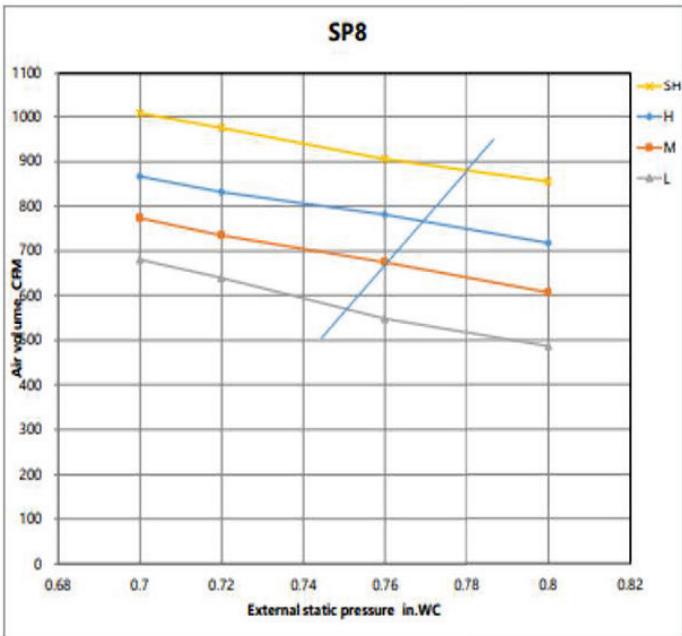
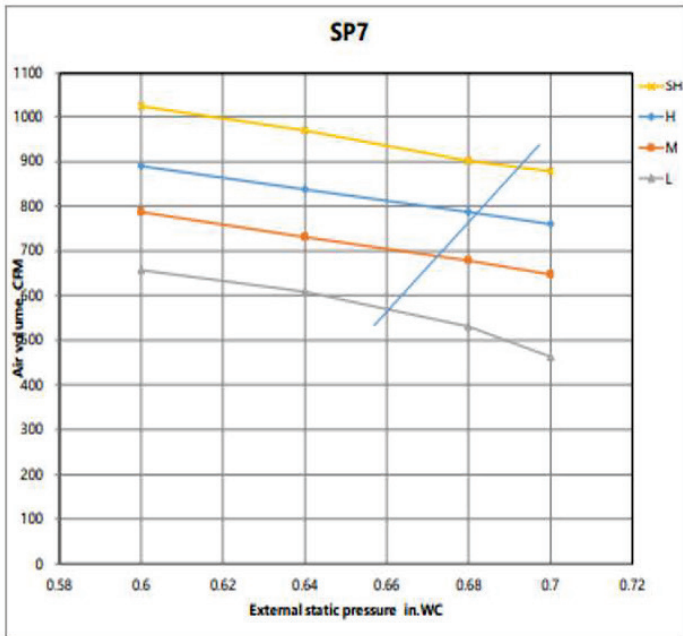
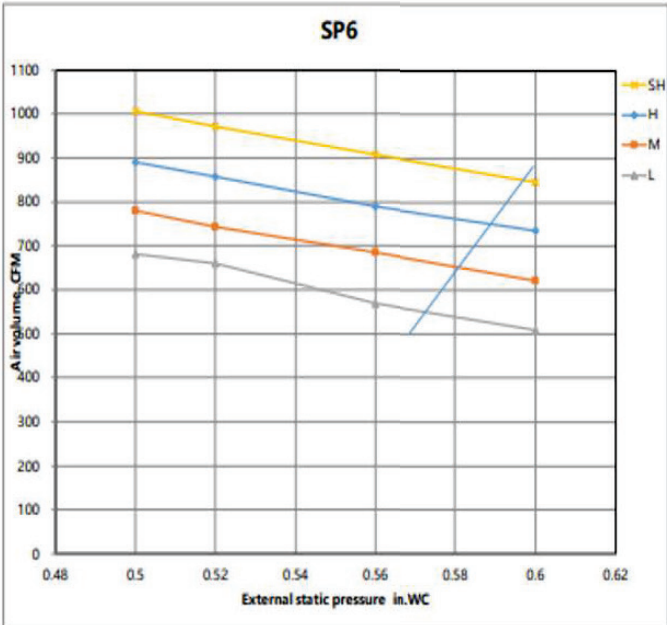
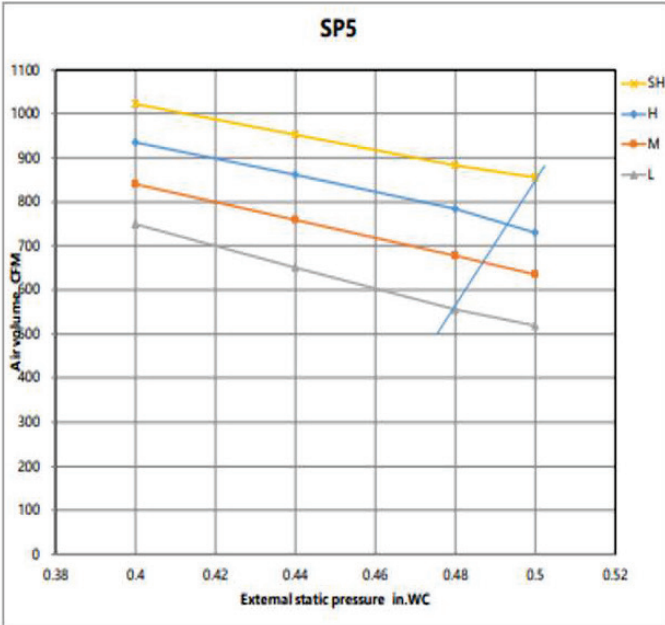
Outdoor Unit Dimensions and Connections



Air Volume

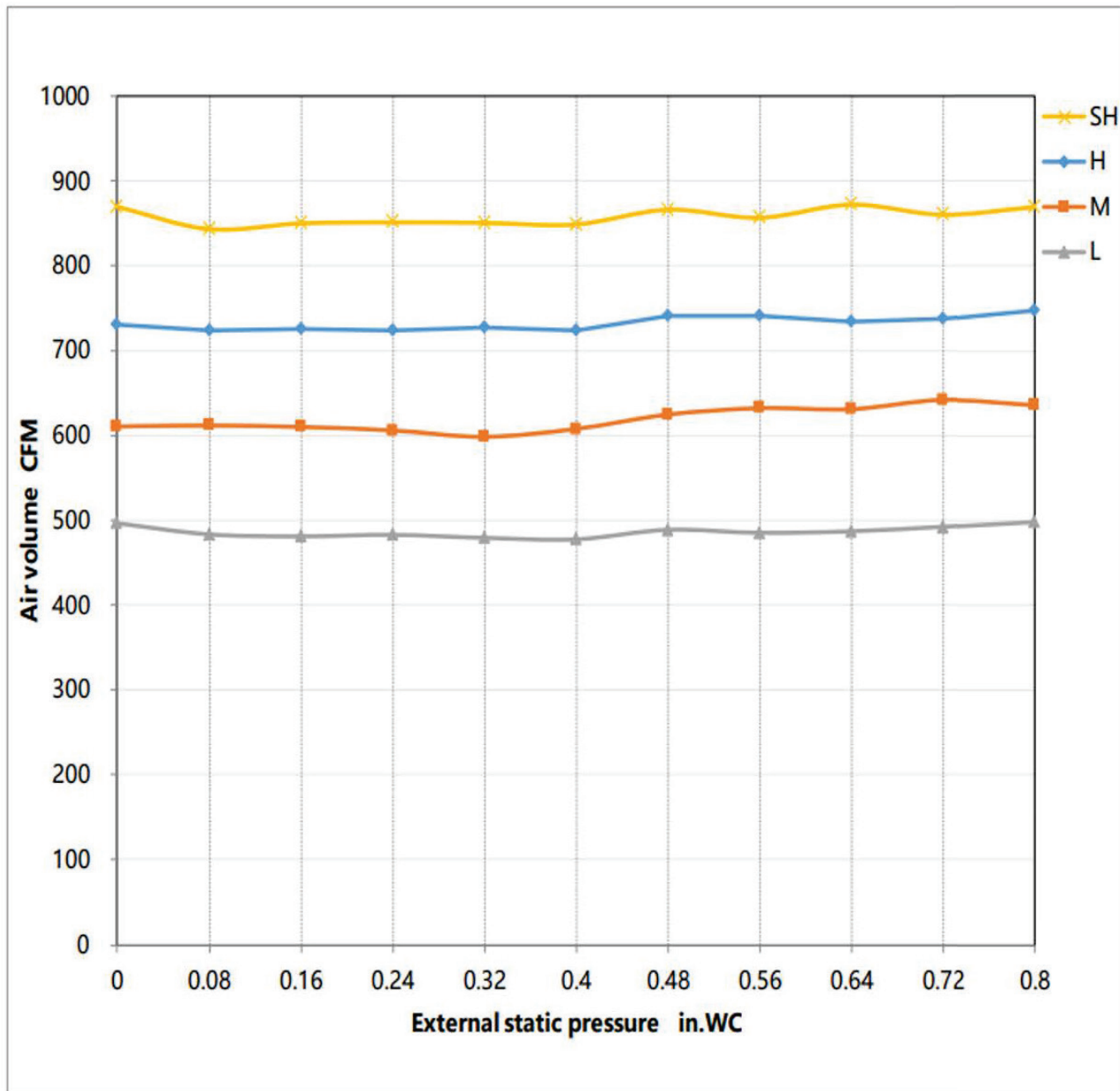


Air Volume



Air Volume

Constant air volume



Climate 5000 Series - Gen 4

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Cooling Performance - Indoor Dry Bulb Temperature at 853 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	20493	20520	20542	20543	20571	21728	21754	21775	21779	21805	23026	23056	23081	23086	23112	24413	24440	24466	24470	24501
	S/T	0.48	0.56	0.63	0.63	0.72	0.50	0.59	0.67	0.67	0.77	0.53	0.63	0.71	0.71	0.81	0.57	0.67	0.76	0.76	0.86
	PI	0.97	0.97	0.97	0.97	0.97	1.02	1.02	1.03	1.03	1.03	1.08	1.09	1.09	1.09	1.09	1.15	1.15	1.15	1.15	1.15
-13°F (-25.0°C)	TC	22201	22230	22254	22255	22285	23538	23566	23590	23594	23622	24945	24977	25005	25010	25038	26448	26476	26505	26509	26542
	S/T	0.48	0.56	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.77	0.54	0.63	0.72	0.72	0.82	0.57	0.67	0.76	0.76	0.87
	PI	1.05	1.05	1.05	1.05	1.05	1.11	1.11	1.11	1.11	1.11	1.18	1.18	1.18	1.18	1.18	1.25	1.25	1.25	1.25	1.25
-4°F (-20.0°C)	TC	23908	23940	23966	23967	24000	25349	25379	25404	25409	25439	26864	26898	26928	26933	26963	28482	28513	28544	28549	28584
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.82	0.57	0.67	0.77	0.77	0.87
	PI	1.13	1.13	1.13	1.13	1.13	1.19	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.27	1.34	1.34	1.34	1.34	1.35
0°F (-17.8°C)	TC	24667	24700	24727	24728	24761	26154	26185	26211	26216	26247	27716	27752	27783	27788	27819	29386	29418	29450	29455	29492
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.16	1.16	1.16	1.16	1.17	1.23	1.23	1.23	1.24	1.24	1.31	1.31	1.31	1.31	1.31	1.38	1.39	1.39	1.39	1.39
5°F (-15.0°C)	TC	25616	25650	25678	25679	25714	27160	27192	27219	27224	27256	28782	28820	28852	28857	28889	30517	30550	30583	30588	30626
	S/T	0.49	0.57	0.65	0.65	0.74	0.51	0.60	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.21	1.21	1.21	1.21	1.21	1.28	1.28	1.28	1.28	1.28	1.36	1.36	1.36	1.36	1.36	1.44	1.44	1.44	1.44	1.44
14°F (-10.0°C)	TC	25625	25660	25687	25688	25723	27169	27202	27228	27234	27266	28793	28830	28862	28867	28900	30527	30561	30594	30599	30637
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.10	1.11	1.11	1.11	1.11	1.17	1.17	1.17	1.17	1.17	1.24	1.24	1.24	1.24	1.25	1.32	1.32	1.32	1.32	1.32
17°F (-8.3°C)	TC	25628	25663	25690	25691	25726	27173	27205	27232	27237	27269	28796	28833	28866	28871	28903	30531	30564	30597	30602	30640
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.79	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.78	0.78	0.88
	PI	1.07	1.07	1.07	1.07	1.07	1.13	1.14	1.14	1.14	1.14	1.20	1.20	1.21	1.21	1.21	1.27	1.28	1.28	1.28	1.28
23°F (-5.0°C)	TC	25627	25662	25689	25690	25725	27172	27204	27231	27236	27268	28795	28832	28864	28870	28902	30530	30563	30596	30601	30639
	S/T	0.49	0.58	0.65	0.65	0.75	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.58	0.69	0.78	0.78	0.89
	PI	1.02	1.02	1.02	1.02	1.02	1.08	1.08	1.08	1.08	1.08	1.14	1.15	1.15	1.15	1.15	1.21	1.22	1.22	1.22	1.22
35°F (1.7°C)	TC	25625	25660	25687	25688	25723	27169	27202	27228	27234	27266	28793	28830	28862	28867	28900	30527	30561	30594	30599	30637
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	0.92	0.92	0.92	0.92	0.92	0.97	0.97	0.97	0.97	0.98	1.03	1.03	1.03	1.03	1.03	1.09	1.09	1.09	1.10	1.10
41°F (5.0°C)	TC	27094	27131	27160	27161	27198	28727	28761	28790	28795	28829	30443	30483	30517	30523	30557	32278	32313	32348	32353	32393
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.20	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.27	1.34	1.35	1.35	1.35	1.35	1.43	1.43	1.43	1.43	1.43
47°F (8.3°C)	TC	28564	28602	28633	28634	28673	30285	30321	30351	30357	30393	32094	32136	32172	32178	32214	34028	34065	34102	34107	34150
	S/T	0.50	0.58	0.66	0.66	0.75	0.53	0.62	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.85	0.59	0.69	0.79	0.79	0.90
	PI	1.48	1.48	1.48	1.48	1.48	1.57	1.57	1.57	1.57	1.57	1.66	1.66	1.66	1.66	1.67	1.76	1.76	1.76	1.76	1.77
50°F (10.0°C)	TC	28561	28599	28630	28631	28670	30282	30318	30348	30354	30390	32091	32133	32169	32175	32211	34025	34062	34099	34104	34147
	S/T	0.50	0.58	0.66	0.66	0.76	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90
	PI	1.51	1.51	1.51	1.51	1.51	1.60	1.60	1.60	1.60	1.60	1.69	1.70	1.70	1.70	1.70	1.80	1.80	1.80	1.80	1.80
59°F (15.0°C)	TC	28553	28591	28622	28623	28662	30273	30309	30339	30345	30381	32082	32124	32159	32165	32201	34015	34052	34089	34094	34137
	S/T	0.50	0.59	0.67	0.67	0.76	0.53	0.62	0.71	0.71	0.81	0.56	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91
	PI	1.60	1.60	1.61	1.61	1.61	1.70	1.70	1.70	1.70	1.70	1.80	1.80	1.80	1.80	1.81	1.91	1.91	1.91	1.91	1.92
68°F (20.0°C)	TC	28544	28583	28613	28615	28653	30264	30300	30330	30336	30372	32073	32114	32150	32156	32192	34005	34042	34079	34084	34127
	S/T	0.51	0.60	0.69	0.69	0.78	0.54	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.61	0.72	0.82	0.82	0.93
	PI	1.70	1.70	1.70	1.70	1.70	1.80	1.80	1.80	1.80	1.81	1.91	1.91	1.91	1.91	1.91	2.02	2.02	2.03	2.03	2.03
77°F (25.0°C)	TC	28536	28574	28605	28606	28645	30256	30292	30321	30327	30363	32063	32105	32141	32147	32183	33995	34032	34069	34075	34117
	S/T	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	0.62	0.73	0.83	0.83	0.94
	PI	1.79	1.79	1.80	1.80	1.80	1.90	1.90	1.90	1.90	1.91	2.01	2.01	2.02	2.02	2.02	2.13	2.14	2.14	2.14	2.14
82°F (27.8°C)	TC	28038	28075	28106	28107	28145	29727	29762	29792	29798	29833	31503	31544	31579	31585	31620	33401	33438	33474	33479	33521
	S/T	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	0.63	0.74	0.84	0.84	0.95
	PI	1.86	1.87	1.87	1.87	1.87	1.98	1.98	1.98	1.98	1.98	2.09	2.10	2.10	2.10	2.10	2.22	2.22	2.23	2.23	2.23
86°F (30.0°C)	TC	27016	27052	27081	27082	27119	28644	28678	28706	28712	28746	30355	30394	30428	30434	30468	32184	32219	32254	32259	32293
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	1.95	1.95	1.95	1.95	1.96	2.07	2.07	2.07	2.07	2.07	2.19	2.19	2.20	2.20	2.20	2.32	2.33	2.33	2.33	2.33
95°F (35.0°C)	TC	24716	24750	24776	24777	24811	26206	2623723													

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Heating Performance - Indoor Dry Bulb Temperature at 853 CFM										
Outdoor DB(°F)	TC□Total Heating Capacity(BTU/hr)					PI□Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	11834	11597	11539	11493	11390	2.49	2.51	2.52	2.52	2.53
-13°F(-25.0°C)	16009	15688	15610	15547	15408	2.68	2.71	2.71	2.72	2.73
-4°F(-20.0°C)	19833	19436	19339	19262	19089	2.87	2.89	2.90	2.90	2.92
0°F(-17.8°C)	22264	21819	21709	21623	21428	2.99	3.01	3.02	3.03	3.04
5°F(-15.0°C)	25302	24796	24672	24573	24352	3.14	3.17	3.17	3.18	3.19
14°F(-10.0°C)	24969	24470	24347	24250	24032	2.90	2.92	2.93	2.93	2.94
17°F(-8.3°C)	24858	24361	24239	24142	23925	2.81	2.84	2.84	2.85	2.86
22°F(-5.6°C)	26138	25615	25487	25385	25157	2.82	2.84	2.85	2.86	2.87
27°F(-2.8°C)	27418	26870	26735	26628	26389	2.83	2.85	2.86	2.86	2.87
32°F(0.0°C)	28698	28124	27984	27872	27621	2.84	2.86	2.86	2.87	2.88
35°F(1.7°C)	29466	28877	28732	28618	28360	2.84	2.86	2.87	2.87	2.89
37°F(2.8°C)	29919	29320	29174	29057	28796	2.78	2.80	2.80	2.81	2.82
42°F(5.6°C)	31050	30429	30277	30156	29884	2.62	2.64	2.64	2.65	2.66
44.6°F(7.0°C)	31503	30873	30718	30595	30320	2.55	2.58	2.58	2.59	2.60
47°F(8.3°C)	32181	31538	31380	31255	30973	2.46	2.48	2.48	2.49	2.50
52°F(11.1°C)	32216	31572	31414	31289	31007	2.26	2.28	2.28	2.28	2.29
57°F(13.9°C)	32251	31606	31448	31323	31041	2.05	2.07	2.07	2.08	2.09
59°F(15.0°C)	32265	31620	31462	31336	31054	1.97	1.99	1.99	2.00	2.00
62°F(16.7°C)	32286	31641	31483	31357	31074	1.85	1.87	1.87	1.87	1.88
64.4°F(18.0°C)	32300	31654	31496	31370	31088	1.77	1.78	1.79	1.79	1.80

Bolded values were tested. All other values are simulated and for reference only.